PALAR: Estimation of Absolute Abundance Effects in Regression with Relative Abundance Predictors

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Abstract

High-dimensional compositional data are ubiquitous in omics research. Microbiome sequencing experiments measure the relative abundances (proportions) of microbial features, while the absolute abundances within the ecosystem remain unobserved. Most regression methods with microbial relative abundance predictors rely on log-ratio transformations and typically impose sparsity on the regression coefficients to manage high dimensionality. However, we show that the sparsity assumption often does not hold for these coefficients. To address this issue, we systematically investigate the relationship between the log-ratio regression and the absolute abundance regression. Motivated by the connection, we propose PALAR, a simple and efficient approach for estimating sparse absolute abundance effects using penalized regression with predictors derived from a new compositional transformation. We applied PALAR to four microbiome studies on colorectal cancer, demonstrating its advantages over existing methods in consistently identifying disease-relevant microbial species and improving prediction accuracy and generalizability.

Keywords: Coefficient sparsity, Log-ratio transformation, Microbiome, Penalized regression, Variable selection

1 Introduction

Compositional data, which represent the relative abundances (RA) of components, are prevalent across numerous disciplines, including biology, economics, and ecology. In recent years, high-dimensional compositional datasets have become increasingly common, particularly in omics fields such as microbiome, metabolomics, and proteomics, where relative quantities are often easier to measure than their absolute counterparts (Greenacre et al., 2021). Notably, microbiome data generated through high-throughput sequencing consist of sequence read counts assigned to hundreds of microbial features (at the microbe or gene level), which are systematically distorted from their true absolute abundances (AA) within a unit volume of the ecosystem. Since sequencing depth is uncorrelated with the actual microbial load (i.e., the total absolute abundance across all features), the observed counts reflect RA (Gloor et al., 2016, 2017) and are routinely normalized to proportions. Although experimental techniques exist to recover AA microbiome data (Fredricks et al., 2007; Ryu et al., 2013; Vandeputte et al., 2017), they remain costly and their accuracy is subject to debate (McLaren et al., 2019).

In this paper, we focus on regression analysis with RA predictors. In high-dimensional settings, penalized regression is widely used to achieve a balance between predictive accuracy, interpretability, and parsimony. Sparsity is typically induced by applying regularization penalties to the regression coefficients (Tibshirani, 1996; Zhu et al., 2022). However, the inherent constraint that RAs must sum to a fixed constant renders traditional penalized regression methods inapplicable (Aitchison, 1982). Figure 1 illustrates that AA and RA effects have distinct sparsity profiles. Because a change in the AA of even a single feature alters the RA of all features, sparsity in RA effects is rarely observed (Figure 1a). A sparse RA effect profile emerges only when the effects of a small subset of features cancel each other out. For example, in the right panels of Figure 1b,c, the effects of features A and B balance each other out without involving other features. In this scenario, a corresponding sparse AA effect profile always exists (Figure 1b left panel). Naturally, the underlying AA effect profile is not unique due to the information loss inherent in RA data. Although alternative AA effect profiles exist, they are generally not sparse (Figure 1c left panel). The illustration hints at several points that motivate our method: (1) the sparsity assumption on RA effects is generally unrealistic; (2) the sparsity assumption on AA effects is more plausible, interpretable, and of fundamental interest in many applications; and (3) sparse AA effects appear to be approximately identifiable from RA data.

Log-ratio regression is the prevailing approach for regression analysis with RA predictors. In this approach, a log-ratio transformation is applied to the composition to map it from the constrained simplex space to Euclidean space, enabling the use of standard regression techniques. The earliest form of log-ratio regression is based on the additive log-ratio (ALR) transformation (Aitchison and Bacon-shone, 1984), with the linear predictor: $\beta_1^* \log(X_1/X_p) + \ldots + \beta_{p-1}^* \log(X_{p-1}/X_p)$, where $(X_1, \ldots, X_p)$ is the vector of RAs for p features, and the last feature serves as the reference. To avoid the arbitrariness of selecting a reference component, the log-contrast model (LCM) offers a symmetric alternative: $\beta_1' \log(X_1) + \ldots + \beta_p' \log(X_p)$, subject to the zero-sum constraint $\sum_{j=1}^p \beta_j' = 0$. Lin et al. (2014) introduced a variable selection approach for LCM by imposing an ℓ_1 penalty on the coefficients β_j''s. The centered log-ratio transformation is also commonly used, and studies have shown that it is equivalent to the LCM when the zero-sum constraint is imposed on the coefficients (Wang and Zhao, 2017; Randolph et al., 2018). Another line of work considers all pairwise log-ratios (Gordon-Rodriguez et al., 2022; Greenacre, 2019; Bates and Tibshirani, 2019; Rivera-Pinto et al., 2018), but these approaches often suffer from computational inefficiencies and identifiability issues.

In this article, we systematically investigate the connection between log-ratio regression and the underlying AA regression, based on the fact that the feature space of any

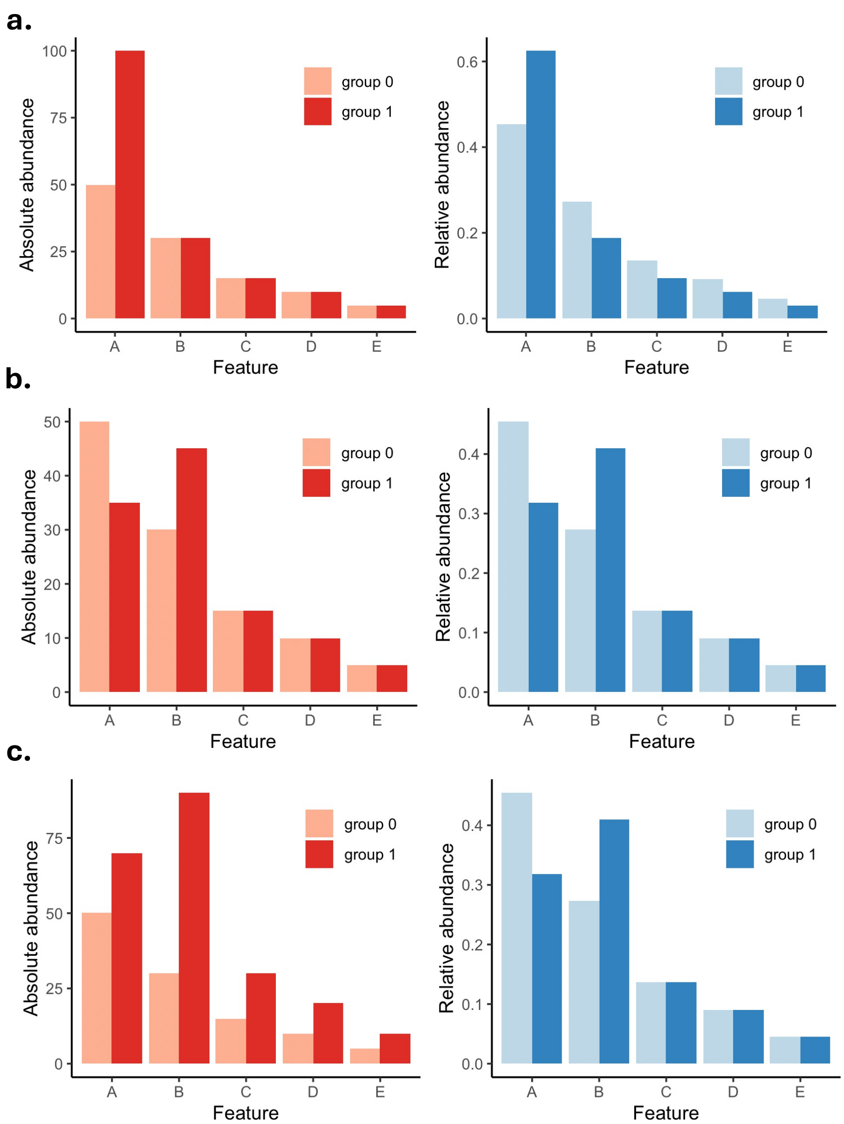

Figure 1: Comparison of effect profiles of absolute abundance (AA, left panel) and relative abundance (RA, right panel) predictors in different scenarios **a**-**c**. In **a**, only Feature A has differential AA between comparison groups; in this scenario, the AA effect profile is sparse, but the RA effect profile is not. The RA effect profiles in **b** and **c** are identical and exhibit a sparse pattern because the changes in Features A and B cancel each other out. This may arise from sparse AA effects (**b**, left) or non-sparse AA effects (**c**, left).

log-ratio-transformed composition lies in a $(p-1)$-dimensional subspace formed by linear combinations of the log-transformed AA variables, representing a 1-dimensional reduction from the original space. We elucidate the relationship between AA effects in AA regression and RA effects in log-ratio regression, demonstrating that RA effects are generally non-sparse, even when the underlying AA effects are sparse. We also characterize the information loss caused by this one-dimensional reduction and examine its impact on model compatibility and prediction accuracy. Motivated by the connection between AA and RA effects, we introduce a new compositional transformation and develop PALAR (Penalized AA effect estimation based on a transformed Log-rAtio Regression), a penalized regression framework that utilizes predictors generated from this new transformation. Under PALAR, we establish the identifiability of sparse AA effects, derive estimation error bounds, and prove variable selection consistency. We demonstrate the advantages of PALAR over existing methods across a wide range of simulation scenarios and apply it to our motivating microbiome datasets of colorectal cancer (as detailed in the next section). Compared to existing methods, PALAR more consistently estimates microbial associations with colorectal cancer across studies, reliably identifies meaningful microbial signatures, and improves colorectal cancer classification accuracy and generalizability.

The remainder of this article is structured as follows. Section 2 introduces motivating datasets and their related scientific questions. Section 3 establishes the connection between log-ratio regression and AA regression, derives the relationship between their coefficients, and characterizes the information loss inherent in log-ratio regression. Section 4 presents the PALAR methodology and establishes theoretical results. Simulation studies and applications to the motivating datasets are presented in Sections 5 and 6, respectively. Section 7 provides concluding discussion. Proofs and additional results are provided in the supplementary materials.

2 Motivating datasets and scientific questions

Colorectal cancer (CRC) remains one of the leading causes of cancer-related morbidity and mortality worldwide (Sung et al., 2021). In recent years, the study of the gut microbiome has emerged as a promising frontier in CRC research and medicine. The gut microbiome plays a critical role in modulating the immune system, metabolizing dietary compounds, and influencing inflammation—all of which are key factors in colorectal carcinogenesis (Brennan and Garrett, 2016). Our motivating datasets are from four microbiome studies of CRC involving subjects from different countries (Wirbel et al., 2019; Yu et al., 2017; Vogtmann et al., 2016; Zeller et al., 2014). Studies 1-4 had sample sizes of 127 (73 cases, 54 controls), 120 (60 cases, 60 controls), 113 (52 cases, 61 controls), and 103 (51 cases, 52 controls). All microbiome data were generated using metagenomic sequencing. Raw sequencing data were reprocessed using a unified bioinformatics pipeline for taxonomic profiling, as previously described (Wirbel et al., 2019). The reprocessed data were obtained from `https://github.com/zellerlab/crc_meta`. After filtering out microbial species with less than 20% prevalence across all samples, 401 species were retained for analysis. We transformed read counts into proportions, replacing zero counts with the maximum rounding error of 0.5 (Aitchison, 1986).

Identifying microbial signatures associated with CRC risk and assessing their predictive value is of scientific interest. However, accurate and reliable signature identification is complicated by the compositional nature of the microbiome data. RA does not provide information about the absolute number of microbial cells. In CRC studies, where shifts in microbial populations may be subtle yet biologically meaningful, using RA data could obscure true AA-level associations with disease, leading to inaccurate signature detection and poor replicability and generalizability across cohorts and populations.

3 Bridging log-ratio regression to AA regression

Suppose the data consist of n independent samples. For sample $i = 1, \ldots, n$, let Y_i denote the continuous outcome variable, $\boldsymbol{X}_i = (X_{i1}, \ldots, X_{ip})$ denote the p-dimensional vector of log-transformed AA variables, $\boldsymbol{X}_i^* = (X_{i1}^*, \ldots, X_{ip-1}^*)$ denote the $(p-1)$-dimensional vector of log-ratio-transformed RA variables. The AA and log-ratio linear regression models are expressed as

$$\text{AA linear regression: } Y_i = \alpha + \boldsymbol{X}_i \boldsymbol{\beta} + \epsilon_i, \tag{1}$$

$$\text{Log-ratio linear regression: } Y_i = \alpha^* + \boldsymbol{X}_i^* \boldsymbol{\beta}^* + \epsilon_i^*, \tag{2}$$

where $\boldsymbol{\beta}$ and $\boldsymbol{\beta}^*$ are the regression coefficients (referred to as AA and RA effects), α and α^* are the intercept coefficients, and ϵ_i and ϵ_i^* are zero-mean normally distributed noise.

For any type of log-ratio transformation, $\boldsymbol{X}_i^*$ can be expressed as a linear projection of $\boldsymbol{X}_i$ onto a $(p-1)$-dimensional subspace. That is, there exists a $p \times (p-1)$ transformation matrix $\boldsymbol{R}$ such that $\boldsymbol{X}_i^* = \boldsymbol{X}_i \boldsymbol{R}$. Based on this connection, we examine the relationship between the two models.

3.1 Relationship between AA and RA effects

Let $\boldsymbol{X} = (\boldsymbol{X}_1^\top, \ldots, \boldsymbol{X}_n^\top)^\top$. The linear predictors in the two models have a relationship

$$\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^* \boldsymbol{\beta}^* + \boldsymbol{\xi}, \tag{3}$$

where $\boldsymbol{\xi}$ lies in the orthogonal complement of the column space of $\boldsymbol{X}^*$.

Since $\boldsymbol{X}^* = \boldsymbol{X}\boldsymbol{R}$, we can express $\boldsymbol{\beta}^*$ as

$$\boldsymbol{\beta}^* = (\boldsymbol{X}^{*\top} \boldsymbol{X}^*)^{-1} \boldsymbol{X}^{*\top} \boldsymbol{X}\boldsymbol{\beta} = (\boldsymbol{R}^\top \boldsymbol{\Sigma}_X \boldsymbol{R})^{-1} \boldsymbol{R}^\top \boldsymbol{\Sigma}_X \boldsymbol{\beta}, \tag{4}$$

where $\boldsymbol{\Sigma}_X$ is the covariance matrix of $\boldsymbol{X}$. Clearly, $\boldsymbol{\beta}^*$ generally lacks sparsity, even when $\boldsymbol{\beta}$ is sparse. Next, we take the ALR transformation as a concrete example to further illustrate the relationship between $\boldsymbol{\beta}$ and $\boldsymbol{\beta}^*$.

Proposition 3.1. *In ALR regression, suppose the pth feature is used as the reference. Assuming this reference is uncorrelated with the other features in the AA space, $\boldsymbol{\beta}^*$ takes the form*

$$\boldsymbol{\beta}^* = \boldsymbol{\beta}_{\setminus p} - C\boldsymbol{\Sigma}_{X_{\setminus p}}^{-1}\mathbf{1}_{p-1},$$

$$C = \frac{\mathbf{1}_p^\top \boldsymbol{\beta}}{\mathbf{1}_p^\top \boldsymbol{\Sigma}_X^{-1}\mathbf{1}_p},$$

where $\boldsymbol{\beta}_{\setminus p}$ denotes the subvector of $\boldsymbol{\beta}$ excluding its pth element, $\boldsymbol{\Sigma}_{X_{\setminus p}}$ denotes the submatrix of $\boldsymbol{\Sigma}_X$ removing the pth row and column, and $\mathbf{1}_p$ denotes the p-dimensional vector of ones.

The proof of Proposition 3.1 is provided in the supplementary materials. Note that the scalar C is invariant to the choice of reference feature and increases with the sum of the AA effects $\mathbf{1}_p^\top\boldsymbol{\beta}$, which tends to be large when most active features share the same direction of effect. The term $C\boldsymbol{\Sigma}_{X_{\setminus p}}^{-1}\mathbf{1}_{p-1}$ depends on the choice of the reference and induces a systematic difference between $\boldsymbol{\beta}$ and $\boldsymbol{\beta}^*$.

3.2 Information loss

The term $\boldsymbol{\xi}$ in (3) represents the inherent information loss in log-ratio regression relative to the AA regression. The following proposition outlines several properties of $\boldsymbol{\xi}$.

Proposition 3.2. *The information loss $\boldsymbol{\xi}$ satisfies the following properties:*

(a) $\boldsymbol{\xi}$ is independent of the linear predictor in log-ratio regression and invariant to the type of log-ratio transformation.

(b) If $\boldsymbol{X}$ follows a multivariate normal distribution, then $\boldsymbol{\xi}$ is also normally distributed with mean $E(\boldsymbol{X})\boldsymbol{Q}_R\boldsymbol{\beta}$ and variance $\boldsymbol{\beta}^\top\boldsymbol{\Sigma}_X\boldsymbol{Q}_R\boldsymbol{\beta}$, where $E(\boldsymbol{X})$ is the mean of $\boldsymbol{X}$ and $\boldsymbol{Q}_R =$

$\boldsymbol{I}_p - \boldsymbol{R}(\boldsymbol{R}^\top \boldsymbol{\Sigma}_X \boldsymbol{R})^{-1} \boldsymbol{R}^\top \boldsymbol{\Sigma}_X$.

(c) $\boldsymbol{\xi} \approx (\mathbf{1}_p^\top \boldsymbol{\beta})(\boldsymbol{X}\mathbf{1}_p^\top / p)$

The proof of Proposition 3.2 is provided in the supplementary materials. If $\boldsymbol{X}$ is multivariate normally distributed, the log-ratio linear regression (2) is compatible with the AA linear regression (1), in the sense that the mean of the information loss is absorbed into the intercept α^*, and the centered information loss is captured in the residual ϵ_i^*. From part (c) of the proposition, we know that the information loss increases with the sum of the AA effects. When the sum of the AA effects is zero, the log-ratio regression incurs no information loss relative to the AA regression. Figure 2 displays simulation results comparing the prediction errors of the two regressions. As expected, the mean squared error (MSE) of the log-ratio regression increases quadratically as the imbalance between positive and negative AA effects grows. It is important to emphasize that it is not possible to recover this information loss and close the prediction performance gap between AA and RA predictors. Our focus, therefore, is on efficiently estimating the sparse AA effects and maximizing prediction accuracy within the RA-based log-ratio regression framework, intending to approach the performance represented by the dashed line in Figure 2.

Beyond linear regression, information loss renders the regular log-ratio model incompatible with the AA model under generalized linear models (GLM). A log-ratio model that is compatible with the AA regression should take the form

$$g[E(Y_i \mid \boldsymbol{X}_i^*, \xi_i)] = \alpha^* + \boldsymbol{X}_i^* \boldsymbol{\beta}^* + \xi_i, \tag{5}$$

where g is the link function. According to Proposition (3.2), ξ_i represents the centered information loss, which follows a zero-mean normal distribution with variance $\sigma_\xi^2(\boldsymbol{\beta}) = \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_X \boldsymbol{Q}_R \boldsymbol{\beta}$, and the intercept α^* absorbs the mean of the information loss. Since ξ_i is

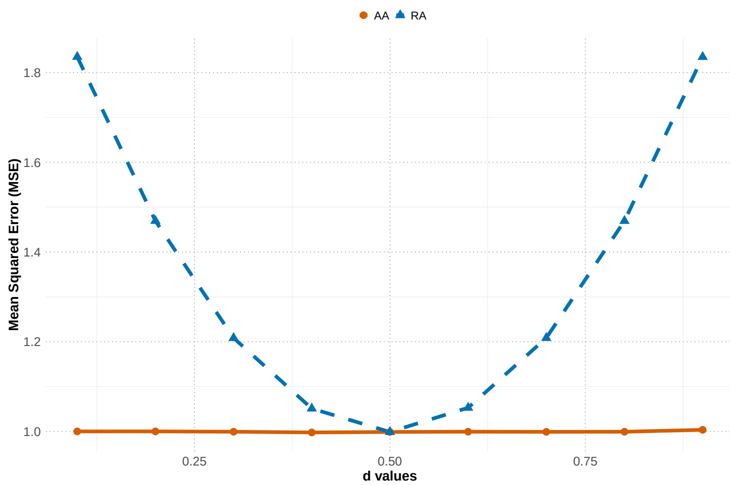

Figure 2: Comparison of prediction errors (measured by mean squared error, MSE) between the AA linear model and the RA log-ratio linear model in simulation studies. The value of d indicates the proportion of negative effects. Linear predictors are computed using the true effects, with the magnitude of nonzero effects fixed at 1. For each value of d, the reported MSE is averaged over 100 simulated datasets.

unobserved, the marginal density of Y_i is defined as

$$f(Y_i \mid \boldsymbol{X}_i^*) = \int_{-\infty}^{\infty} f(Y_i \mid \boldsymbol{X}_i^*, \xi_i)\phi(\xi_i)\, d\xi_i,$$

where ϕ is the density function of a zero-mean normal distribution with variance $\sigma_\xi^2(\boldsymbol{\beta})$. In general, the marginal density does not have a closed-form expression and must be approximated numerically. Iterative algorithms commonly used for parameter estimation in mixed-effects models can be adapted to fit this model.

4 Methodology

4.1 PALAR linear regression

As shown in the previous section, the RA effect vector $\boldsymbol{\beta}^*$ is generally not sparse, even when the underlying AA effect vector $\boldsymbol{\beta}$ is sparse. This motivates us to estimate the sparse

$\boldsymbol{\beta}$ directly instead of $\boldsymbol{\beta}^*$. To this end, we introduce a new compositional transformation

$$\boldsymbol{X}^{**} = \boldsymbol{X}^*(\boldsymbol{R}^\top \boldsymbol{\Sigma}_X \boldsymbol{R})^{-1} \boldsymbol{R}^\top \boldsymbol{\Sigma}_X. \tag{6}$$

Based on equation (4), the linear predictor of the log-ratio regression can be written as $\boldsymbol{X}^{**}\boldsymbol{\beta}$. The transformed variables $\boldsymbol{X}^{**}$ have several nice properties and interpretations:

- $\boldsymbol{X}^{**}$ has the highest overall correlation with $\boldsymbol{X}$, defined as the trace of the covariance $\mathrm{Cov}(\boldsymbol{X}, \boldsymbol{X}^{**})$, among all linear transformations of $\boldsymbol{X}^*$.
- $\boldsymbol{X}^{**}$ is invariant to the choice of log-ratio transformation. That is, for any two log-ratio transformation matrices $\boldsymbol{R}_1$ and $\boldsymbol{R}_2$, we obtain the same result: $\boldsymbol{X}^{**} = \boldsymbol{X}\boldsymbol{R}_1(\boldsymbol{R}_1^\top \boldsymbol{\Sigma}_X \boldsymbol{R}_1)^{-1} \boldsymbol{R}_1^\top \boldsymbol{\Sigma}_X = \boldsymbol{X}\boldsymbol{R}_2(\boldsymbol{R}_2^\top \boldsymbol{\Sigma}_X \boldsymbol{R}_2)^{-1} \boldsymbol{R}_2^\top \boldsymbol{\Sigma}_X$.
- $\boldsymbol{X}^{**}$ recovers the AA effects in the linear predictor. Specifically, for any AA effect vector $\boldsymbol{\beta}$, the linear predictors based on $\boldsymbol{X}$ and $\boldsymbol{X}^{**}$ satisfy: $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^{**}\boldsymbol{\beta} + \boldsymbol{\xi}$. In contrast, the predictor effect in the log-ratio regression is different from the AA effects: $\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^*\boldsymbol{\beta}^* + \boldsymbol{\xi}$ and $\boldsymbol{\beta}^* \neq \boldsymbol{\beta}$.

The following theorem establishes the condition under which sparse $\boldsymbol{\beta}$ is identifiable using Lasso regression with predictors $\boldsymbol{X}^{**}$.

Theorem 4.1. *The S-sparse AA effect vector $\boldsymbol{\beta}$ is identifiable if the Restricted Isometry Property (RIP) constant of order $2s$ satisfies $\delta_{2s}(\boldsymbol{X}^{**}) < 1/3$.*

The condition mirrors that of Proposition 7.11 of Wainwright (2019). Although $\boldsymbol{X}^{**}$ is a projection of $\boldsymbol{X}$ onto a $(p-1)$-dimensional space, its compatibility conditions (e.g., RIP) are generally preserved. As a result, $\boldsymbol{X}^{**}$ remains effective for ensuring the identifiability of $\boldsymbol{\beta}$.

The $\boldsymbol{X}^{**}$ involves the unknown AA covariance matrix $\boldsymbol{\Sigma}_X$. Several methods have been proposed in the literature to estimate $\boldsymbol{\Sigma}_X$ in the context of high-dimensional compositional data. In our numerical studies, we adopt the SparCC method (Friedman and Alm, 2012). Let $\widetilde{\boldsymbol{X}}^{**}$ denote the surrogate of $\boldsymbol{X}^{**}$ obtained by plugging in an estimate of $\boldsymbol{\Sigma}_X$. The PALAR ℓ_1-penalized estimate of $\boldsymbol{\beta}$ is defined as the solution to the optimization problem

$$(\widehat{\alpha}^*, \widehat{\boldsymbol{\beta}}) = \arg\min \left\{ \frac{1}{2n} \sum_{i=1}^{n} (Y_i - \alpha^* - \widetilde{\boldsymbol{X}}_i^{**} \boldsymbol{\beta})^2 + \lambda_n \|\boldsymbol{\beta}\|_1 \right\}. \tag{7}$$

This problem can be efficiently solved using standard Lasso solvers and the tuning parameter λ_n can be selected via cross-validation.

We next present theoretical results on the estimation error bounds and variable selection consistency of the PALAR estimator $\widehat{\boldsymbol{\beta}}$. These results are adapted from standard Lasso theory (Wainwright, 2019), with modifications to account for the approximation error introduced by using the surrogate $\widetilde{\boldsymbol{X}}^{**}$ in place of the true $\boldsymbol{X}^{**}$. Proofs are provided in the supplementary materials. Let $\boldsymbol{Y}$, $\boldsymbol{X}^{**}$, $\widetilde{\boldsymbol{X}}^{**}$, and $\boldsymbol{\epsilon}^*$ denote the stacked vectors or matrices of Y_i, $\boldsymbol{X}_i^{**}$, $\widetilde{\boldsymbol{X}}_i^{**}$, and ϵ_i^*, respectively. Let $\boldsymbol{\beta}^0$ denote the true AA effect vector, S be its support set, $\widetilde{\boldsymbol{w}} = (\boldsymbol{X}^{**} - \widetilde{\boldsymbol{X}}^{**})\boldsymbol{\beta}^0$, and $\widetilde{\boldsymbol{X}}_S^{**}$ denote the submatrix of $\widetilde{\boldsymbol{X}}^{**}$ consisting of the columns corresponding to the indices in the set S. For any vector $\boldsymbol{v}$, let $\|\boldsymbol{v}\|_r$ be the vector ℓ_r-norm. For simplicity, we omit the intercept term in the results below.

Theorem 4.2. *Consider the S-sparse $\boldsymbol{\beta}^0$ in the linear regression model $\boldsymbol{Y} = \boldsymbol{X}^{**}\boldsymbol{\beta} + \boldsymbol{\epsilon}^*$, if the surrogate design matrix $\widetilde{\boldsymbol{X}}^{**}$ satisfies the restricted eigenvalue (RE) condition with parameters $(\kappa, 3)$ and the regularization parameter lower bounded as $\lambda_n \geq 2\|\boldsymbol{X}^{**\top}\boldsymbol{\epsilon}^*/n\|_\infty + 2\|\widetilde{\boldsymbol{X}}^{**\top}\widetilde{\boldsymbol{w}}/n\|_\infty$, the PALAR estimator $\widehat{\boldsymbol{\beta}}$ obtained from (7) satisfies the bound*

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0\|_2 \leq \frac{3}{\kappa}\sqrt{s}\lambda_n.$$

This theorem ensures that the ℓ_2 estimation error of the PALAR Lasso is small. The lower abound on λ_n differs from standard Lasso theory due to the need to account for the difference between $\widetilde{\boldsymbol{X}}^{**}$ and $\boldsymbol{X}^{**}$. As $\widetilde{\boldsymbol{X}}^{**}$ better approximates $\boldsymbol{X}^{**}$, the term $\|\widetilde{\boldsymbol{X}}^{**\top}\widetilde{\boldsymbol{w}}/n\|_\infty$ becomes smaller, thereby reducing the overall estimation error.

Theorem 4.3. *Consider S-sparse $\boldsymbol{\beta}^0$ in the linear regression model $\boldsymbol{Y} = \boldsymbol{X}^{**}\boldsymbol{\beta} + \boldsymbol{\epsilon}^*$. Suppose the surrogate design matrix $\widetilde{\boldsymbol{X}}^{**}$ satisfies the conditions*

(a) The smallest eigenvalue of the Gram matrix satisfies $\gamma_{\min}\left(\frac{\widetilde{\boldsymbol{X}}_S^{**\top}\widetilde{\boldsymbol{X}}_S^{**}}{n}\right) \geq C_{\min} > 0$.

(b) There exists $a \in [0,1]$ such that $\max_{j\in S^c} \|(\widetilde{\boldsymbol{X}}_S^{**\top}\widetilde{\boldsymbol{X}}_S^{**})^{-1}\widetilde{\boldsymbol{X}}_S^{**\top}\widetilde{X}_j^{**}\|_1 \leq a$,

Then for any regularization parameter $\lambda_n \geq \frac{2}{1-a}\left\|\widetilde{\boldsymbol{X}}_{S^c}^{**\top}\Pi_{S^\perp}(\widetilde{\boldsymbol{X}}^{**})\frac{\boldsymbol{\epsilon}^*+\widetilde{\boldsymbol{w}}}{n}\right\|_\infty$, where $\Pi_{S^\perp}(\widetilde{\boldsymbol{X}}_S^{**}) = \boldsymbol{I}_n - \widetilde{\boldsymbol{X}}_S^{**}\left(\widetilde{\boldsymbol{X}}_S^{**\top}\widetilde{\boldsymbol{X}}_S^{**}\right)^{-1}\widetilde{\boldsymbol{X}}_S^{**\top}$, the PALAR Lasso (7) has the following properties:

(1) There is a unique optimal solution $\widehat{\boldsymbol{\beta}}$.

(2) This solution has its support set $\widehat{S}$ contained within the true support set S.

(3) $\|\widehat{\boldsymbol{\beta}}_S - \boldsymbol{\beta}_S^0\|_\infty \leq \left\|\left(\frac{\widetilde{\boldsymbol{X}}_S^{**\top}\widetilde{\boldsymbol{X}}_S^{**}}{n}\right)^{-1}\widetilde{\boldsymbol{X}}_S^{**\top}\frac{\boldsymbol{\epsilon}^*+\widetilde{\boldsymbol{w}}}{n}\right\|_\infty + \left|\!\left|\!\left|\left(\frac{\widetilde{\boldsymbol{X}}_S^{**\top}\widetilde{\boldsymbol{X}}_S^{**}}{n}\right)^{-1}\right|\!\right|\!\right|_\infty \lambda_n$, where $|\!|\!|X|\!|\!|_\infty$ is the matrix ℓ_∞ norm.

(4) No false exclusion if $\min_{i\in S}|\beta_i^0| > B(\lambda_n;\widetilde{\boldsymbol{X}})$, where $B(\lambda_n;\widetilde{\boldsymbol{X}})$ is the ℓ_∞ upper bound in (3).

Property (2) states that PALAR Lasso does not falsely include variables outside the true support of $\boldsymbol{\beta}^0$. Property (4) ensures it does not falsely exclude variables within the support. These properties establish that PALAR Lasso achieves variable selection consistency.

4.2 PALAR generalized linear regression

From Section 3.2, we know that the regular log-ratio GLM and the AA GLM are not compatible. Although it would be desirable to work with the compatible model given in (5), the associated computational burden is non-trivial and will be addressed in future work.

In this paper, we proceed by ignoring the information loss in the model. Specifically, in logistic regression with a binary outcome Y_i, the PALAR ℓ_1-penalized estimate of $\boldsymbol{\beta}$ is defined as the solution to the optimization problem:

$$(\widehat{\alpha}^*, \widehat{\boldsymbol{\beta}}) = \arg\min \left\{ -\frac{1}{n} \sum_{i=1}^{n} \ell_i(\alpha^*, \boldsymbol{\beta}) + \lambda_n \|\boldsymbol{\beta}\|_1 \right\}, \tag{8}$$

$$\text{where} \quad \ell_i(\alpha^*, \boldsymbol{\beta}) = Y_i \log\left(\frac{e^{\alpha^* + \widetilde{\boldsymbol{X}}_i^{**}\boldsymbol{\beta}}}{1 + e^{\alpha^* + \widetilde{\boldsymbol{X}}_i^{**}\boldsymbol{\beta}}} \right) + (1 - Y_i) \log\left(\frac{1}{1 + e^{\alpha^* + \widetilde{\boldsymbol{X}}_i^{**}\boldsymbol{\beta}}} \right).$$

This optimization problem can be efficiently solved using standard Lasso solvers and the tuning parameter λ_n can be selected via cross-validation.

Although the information loss is not explicitly addressed, simulation studies and real data applications show that PALAR substantially outperforms regular log-ratio logistic regression in terms of estimation accuracy, prediction performance, and variable selection.

5 Simulation studies

We conducted simulation studies to evaluate the numerical performance of PALAR using the estimated AA covariance and the true AA covariance (PALAR-TC). We compared these methods with ALR regression, LCM, and the regression model using log-proportion predictors without zero-sum constraint on the regression coefficients (LP). In ALR regression, the reference feature was randomly selected from the p features. LCM was applied only in the continuous outcome setting due to the availability of R code from Lin et al. (2014). All other methods were implemented using the Lasso solver from the glmnet R package. All methods used ten-fold cross-validation to select the tuning parameter.

We generated the realistic compositional data using one of the motivating datasets as the template. Specifically, we first simulated log(AA) data $\boldsymbol{X} = \{x_{ij}\}$ from a multivariate normal distribution $\mathcal{N}_p(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$, and then obtained the proportions $\boldsymbol{P} = \{p_{ij}\}$ using the softmax transformation $p_{ij} = \exp(x_{ij}) / \sum_{k=1}^{p} \exp(x_{ik})$. The parameters $\boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_X$ were specified based on the microbiome template dataset. We retained the top p most abundant microbial features in the template and let $\boldsymbol{P}^{(D)}$ denote the matrix of the observed proportions for those features. To specify the true covariance $\boldsymbol{\Sigma}_X$, we applied SparCC to $\boldsymbol{P}^{(D)}$ to estimate the AA covariance structure. We then constructed the template AA data by multiplying $\boldsymbol{P}^{(D)}$ with a vector of microbial load values (i.e., total AA across features), sampled from a negative binomial distribution with mean 10^6 and variance 2×10^6. The log-transformed values of this constructed AA matrix were used to compute the sample mean, which was then specified as our true $\boldsymbol{\mu}_X$.

Using the simulated $\boldsymbol{X}$, we generated continuous outcomes according to the AA linear regression model and binary outcomes according to the AA logistic regression model. We randomly selected 10% of the features as active features with nonzero AA effects. To evaluate the impact of effect direction imbalance, we considered three scenarios: $d = 0.5$ represents an equal number of positive and negative effects; $d = 0.75$ represents that 75% of the active features have negative effects and 25% have positive effects; $d = 1$ corresponds to all active features having negative effects. The magnitudes of the nonzero effects were sampled from $|N(0, 1)|$.

We considered sample sizes $n = 200, 400, 800$ and feature dimensions $p = 200, 400$, and repeated 100 simulations for each setting. Performance was evaluated using four metrics. Estimation error was measured by the ℓ_1 loss $\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0\|_1$, where $\widehat{\boldsymbol{\beta}}$ denotes the estimated effects. In ALR regression, the coefficient for the reference feature cannot be estimated and is therefore fixed at zero. Prediction performance was assessed using MSE for continuous outcomes and the area under the receiver operating characteristic curve (AUC) for binary outcomes. Both metrics were evaluated on an independent test set of size n. To construct the PALAR predictors $\boldsymbol{X}^{**}$ in the test set, we used the AA covariance matrix $\boldsymbol{\Sigma}_X$ estimated from the training data. We also evaluated variable selection performance using the false positive rate (FPR) and false negative rate (FNR), where positives and negatives refer to features with nonzero and zero AA effects, respectively.

Figure 3 presents the results on estimation and prediction performance, and Figure S1 shows the variable selection performance under different values of d. As d increases, the effect directions become more imbalanced, and the sum of effect sizes deviates further from zero. In the balanced scenario ($d = 0.5$), the performance of all methods is relatively similar. However, as the imbalance in effect directions increases, the advantages of PALAR-TC and PALAR become more pronounced. PALAR-TC consistently outperforms PALAR, as it uses the true covariance matrix. The performance gap narrows with increasing sample size, reflecting improved estimation of the AA covariance matrix. ALR and LP regressions perform similarly across all settings. Compared to ALR and LP regressions, LCM yields more accurate estimation but worse prediction performance, with lower FPR but higher FNR. Figure S2 provides an illustrative example to further highlight the variable selection performance of different methods under varying values of d. The marginal regression coefficients are fluid depending on the form of the predictors. As d increases, ALR and LP regressions and LCM tend to select a large number of null features. In contrast, PALAR achieves more accurate selections and demonstrates robustness to changes in d.

The results of the logistic regression simulations are shown in Figures S3 and S4. PALAR and PALAR-TC continue to outperform the other methods, with their advantages becoming more pronounced as d increases.

6 Data analysis

We applied PALAR to four gut microbiome studies (Wirbel et al., 2019; Yu et al., 2017; Vogtmann et al., 2016; Zeller et al., 2014) to identify gut microbial signatures associated with CRC. As in the simulation studies, we also applied ALR and LP regressions for comparison. For ALR, the reference taxon was randomly selected from the 50 species with relatively high abundance and low variance, measured by the coefficient of variation.

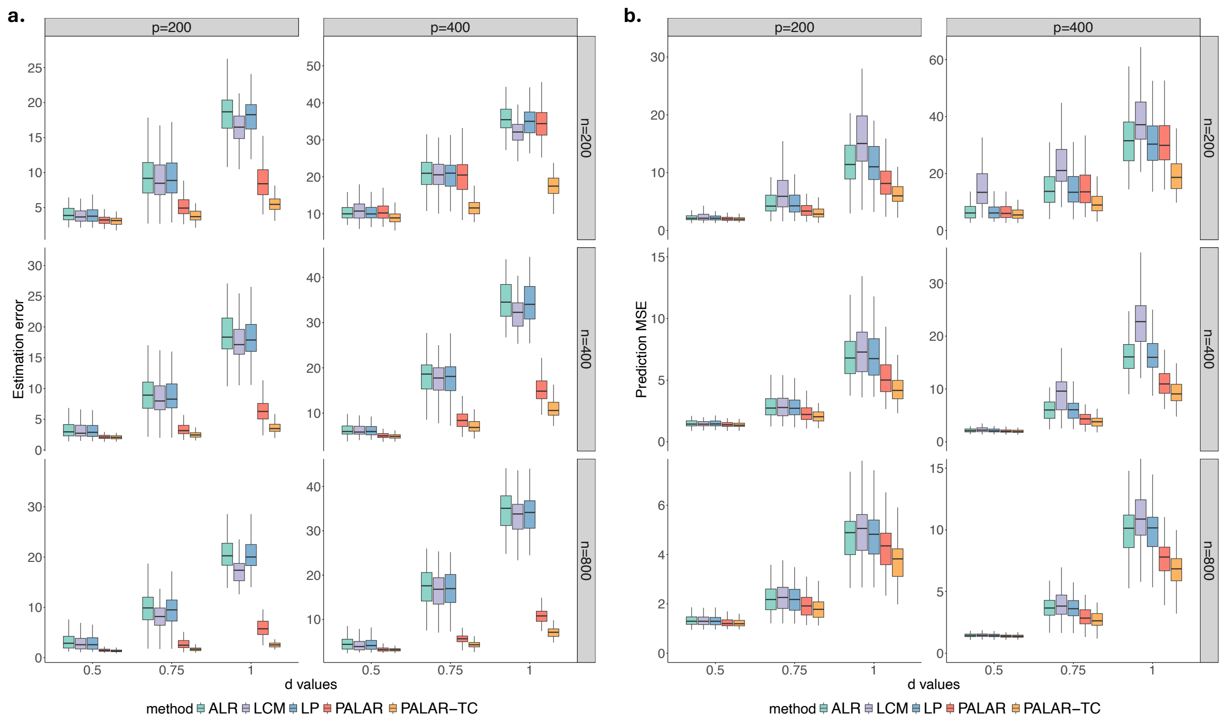

Figure 3: Comparison of different methods based on estimation error (**a**) and prediction error (**b**) in simulations with the continuous outcome under varying d values. The d value is the proportion of active features with negative effects.

We first analyzed each study individually and evaluated the concordance of effect estimates across studies. Specifically, for each pair of studies s and s', we computed the mean squared deviation (MSD) of effect estimates as $\frac{1}{p}\sum_{j=1}^{p}\left(\widehat{\beta}_j^{(s)} - \widehat{\beta}_j^{(s')}\right)^2$, where $\widehat{\beta}_j^{(s)}$ denote the effect estimate for feature j in study s. To ensure robustness, we repeated this evaluation on 100 bootstrap samples of the four studies and reported the average MSD across replicates (Figure 4). Among the methods compared, PALAR achieved the highest consistency of effect estimates across studies, while LP showed the lowest. These results suggest that PALAR is better able to capture stable microbial associations that are less influenced by inter-study variability and technical batch effects commonly present in microbiome data.

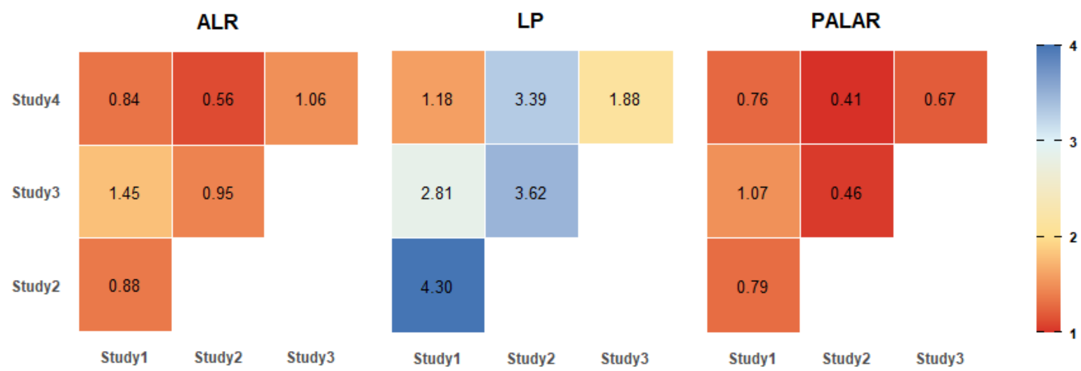

Figure 4: Comparison of methods in effect estimate concordance across studies. The heatmap displays the mean squared deviation of effect estimates between study pairs, with lower values indicating higher concordance.

Next, we identified CRC-associated microbial features by pooling data from all studies to maximize discovery power. To ensure stable feature selection, we generated 100 bootstrap samples of the pooled dataset and applied each method to select species within each sample. Species identified by PALAR in at least 90% of the bootstrap replicates are shown in Table 1, while those selected by ALR and LP regressions are shown in Tables S1 and S2, respectively. Overall, PALAR identified significantly more species enriched in CRC than depleted ones, whereas ALR and LP tended to yield a more balanced number of enriched and depleted species. PALAR also recovered several well-established CRC-associated species, including *Fusobacterium nucleatum*, *Peptostreptococcus stomatis*, *Parvimonas micra*, *Gemella morbillorum*, and *Alistipes onderdonkii*—all of which have been reported as enriched in CRC patients in multiple studies (Reyes et al., 2001; Feng et al., 2015; Hasan et al., 2022). Notably, *Fusobacterium nucleatum*, *Peptostreptococcus stomatis* and *Alistipes onderdonkii* were missed by ALR regression, and *Alistipes onderdonkii* was also missed by LP regression. We further evaluated the out-of-sample CRC prediction performance, measured by AUC and Brier score, using species selected by each method. Within each bootstrap replicate, we randomly split the data into a 90% training set and a 10% testing set. Species selected by PALAR yielded the highest prediction accuracy (Table 2).

Table 1: CRC-associated gut microbial species identified by PALAR. Species listed were selected in 90% of bootstrap replicates. Beta is the refitted effect estimate based on the pooled dataset.

Species	**Beta**
Positive Effects:	
Solobacterium moorei [0531]	0.7004
Coprococcus catus [4874]	0.4954
Peptostreptococcus stomatis [4614]	0.4874
Gemella morbillorum [4513]	0.4623
unknown Ruminococcaceae [7652]	0.4432
unknown Clostridiales [6348]	0.3924
unknown Bacteroidales [5329]	0.3414
Fusobacterium nucleatum [0776]	0.3259
unknown Prevotella [5668]	0.2880
Parvimonas micra [1145]	0.2858
unknown Clostridiales [6009]	0.2831
Veillonella parvula [1042]	0.2702
Subdoligranulum species [4738]	0.2570
Clostridium symbiosum [1475]	0.2212
unknown Anaerotruncus [6835]	0.2153
Alistipes onderdonkii [0775]	0.2042
Ruminococcus species [6664]	0.1779
Acidaminococcus intestini [0391]	0.1454
unknown Firmicutes [6091]	0.1321
Phascolarctobacterium species [2805]	0.1183
unknown Clostridiales [6105]	0.0981
Negative Effects:	
uncultured Eubacterium species [6201]	−1.1068
unknown Clostridiales [6107]	−0.4457
unknown Clostridiales [5880]	−0.3563
unknown Clostridiales [6700]	−0.3259
Clostridium species [5681]	−0.2676
Clostridium species [7788]	−0.2542
unknown Clostridium [5413]	−0.0937

Table 2: Prediction performance on bootstrap pooled data. Average AUC and Brier scores across 100 bootstrap replicates are reported, with standard deviations shown in parentheses.

Method	**AUC**	**Brier**
ALR	0.7965 (0.0587)	0.1835 (0.0239)
LP	0.8016 (0.0585)	0.1815 (0.0241)
PALAR	0.8104 (0.0592)	0.1773 (0.0243)

We further performed the leave-one-study-out (LOSO) prediction to assess generalizability. The data from one study was set aside as an external testing set, while the data from the remaining three studies was pooled as a training set, on which we performed cross-validation for model tuning. We evaluated the performance of the trained model on the held-out study. We repeated the procedure 100 times with different random splits of the cross-validation folds and reported the results in Table 3. Species selected by PALAR consistently achieved the best CRC prediction performance across all held-out studies, demonstrating strong generalizability.

Table 3: Prediction performance under leave-one-study-out (LOSO) evaluation. Average AUC and Brier scores across 100 random splits of the cross-validation folds are reported, with standard deviations shown in parentheses.

Held-out study	**Method**	**AUC**	**Brier**
Study1	ALR	0.8058 (0.0056)	0.1767 (0.0028)
	LP	0.7431 (0.0022)	0.2025 (0.0004)
	PALAR	0.8114 (0.0009)	0.1745 (0.0005)
Study2	ALR	0.7801 (0.0156)	0.1915 (0.0057)
	LP	0.6827 (0.0083)	0.2220 (0.0015)
	PALAR	0.7985 (0.0049)	0.1826 (0.0027)
Study3	ALR	0.7482 (0.0187)	0.2095 (0.0074)
	LP	0.7732 (0.0025)	0.1981 (0.0007)
	PALAR	0.7818 (0.0059)	0.1964 (0.0014)
Study4	ALR	0.6921 (0.0121)	0.2265 (0.0047)
	LP	0.6952 (0.0033)	0.2246 (0.0002)
	PALAR	0.7087 (0.0034)	0.2188 (0.0001)

7 Discussion

Our work is motivated by the analysis of gut microbiome data from CRC studies. A key statistical challenge in such datasets is that microbiome sequencing provides only RA data, while the true biological signals of interest are often encoded in the unobserved AA. Existing regression methods with RA predictors typically rely on log-ratio transformations and impose sparsity assumptions on the coefficients. However, we show that the sparsity assumption often does not hold for these log-ratio regression coefficients, leading to unreliable effect estimation and variable selection. To address this issue, we propose PALAR, a new penalized regression framework that explicitly connects RA-based log-ratio regression to the underlying AA regression. PALAR estimates sparse AA effects using a new compositional transformation that recovers the linear predictor up to an irreducible information loss. We establish theoretical guarantees for identifiability, estimation error, and variable selection consistency.

Extensive simulations demonstrate PALAR's superior estimation, variable selection, and prediction performance, particularly in settings with imbalanced effect directions where existing methods struggle. Analyses of gut microbiome data from CRC studies further demonstrate the advantages of PALAR. The method produced more consistent effect estimates across studies and successfully identified several well-established CRC-associated microbial species that were missed by other methods. The improved concordance of effect estimates and enhanced predictive performance underscore PALAR's potential for uncovering true associations in high-dimensional compositional data.

PALAR offers several avenues for extension and improvement. First, its performance could be enhanced with a more accurate estimation of the AA covariance matrix. Future work may explore more robust and adaptive strategies for estimating this matrix for high-dimensional compositional data. Second, although PALAR has been demonstrated in logistic regression models, the current formulation does not explicitly account for the variation in information loss. Extending PALAR to a generalized linear mixed-effects framework that incorporates this information loss can potentially improve the estimation efficiency. Finally, the current implementation employs the Lasso penalty to induce sparsity. However, PALAR is flexible and agnostic to the choice of regularization. Incorporating phylogenetic tree information into microbiome data analysis has been shown to improve the performance of various statistical methods (Hong et al., 2023; Tang et al., 2017; Wang and Zhao, 2017). Our framework can readily accommodate alternative penalties, including structured sparsity approaches such as tree-guided regularization (Chen et al., 2015).

Supplementary Material

Supplementary Material for "PALAR: Estimation of Absolute Abundance Effects in Regression with Relative Abundance Predictors": A document contains details of the proofs and supplementary results (.pdf)

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